

Optimized YOLO-Based Real-Time Crowd Detection for Children's Playground Safety Monitoring

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Abstract— Crowd monitoring is often overlooked, despite its importance in maintaining a comfortable and safe environment. While such events are typically associated with large public gatherings, crowd crushes can occur in any location and can affect anyone. It is particularly important in safety-critical environments such as children's playgrounds, which are prone to overcrowding during peak times, for example, school holidays. Spatial monitoring becomes crucial to prevent any crowd-related incident that may lead to possible injuries involving children. With the advancement of technologies, crowd monitoring can easily be achieved by implementing a crowd detection system, utilizing computer vision, object detection models, and data analytics tools. Despite the existence of simple human detection models, models specifically designed for children's playground remain limited. This research aims to optimize the use of object detection for real-time crowd monitoring in children's playgrounds. By utilizing YOLO models, this research aims to optimize the models by training them on a self-collected dataset, evaluating each model on performance evaluation metrics, and comparing them on quantitative and qualitative measures to find the best performing model. This paper proposed a YOLOv8-based Playground Crowd Detection model trained on 650 playground images, achieving score for precision of 0.939, recall of 0.922, mAP@0.50 of 0.963. The system aims to aid in playground monitoring, especially for child safety.

Keywords— computer vision, human detection system, safety monitoring system, crowd computing, data analytics

I. INTRODUCTION

Ensuring safety and hazard mitigation in high-density environments requires automated, real-time surveillance to quickly detect overcrowding and spatial anomalies. While manual video monitoring is severely constrained by human cognitive fatigue, the integration of Internet of Things (IoT) frameworks [1], cloud architectures, and web-based analytical management applications [2] has established scalable foundations for automated tracking, risk management, and operational decision support [3]. Recent advances in computational intelligence and machine learning have significantly enhanced automated visual pattern extraction, image analysis[4], and predictive modeling across complex domains [5]. Recent paradigms emphasize modular model architectures [6], domain-knowledge integration in hierarchical learning [7], and hardware acceleration benchmarks [8] to optimize low-latency processing. Addressing these challenges, this paper presents an optimized You Only Look Once (YOLO) deep learning framework

specifically designed for real-time crowd detection and safety monitoring in children's playgrounds, delivering enhanced feature fusion and lightweight execution suitable for edge deployment.

One of the most dangerous situations that can arise in public areas is crowd congestion, which is the condition in which there is a greater number of people in a given area than its own capacity. This makes people not only uncomfortable, but unsafe due to the high density of the area. In spaces with large numbers of people gathering, even a minor mistake can turn tragic. The lack of crowd control can result in crowd crush, potentially causing injuries, fatalities, and panic to rise. A study has shown that there has been an increase in number of crowd crush in recent years, with an average of 50 fatalities per incident [9].

In November 2021, The Astroworld Festival was attended by thousands of fans wanting to see the performance by an American rapper, Travis Scott. The crowd soon turned chaos due to the density and lack of crowd control, which resulted in fans breaking through fences and barriers of the concert area. The crowd of audience worsened when Travis Scott appeared on stage, which then created a wave of uncontrollable movement, resulting in many injuries and ten fatalities [10]. A similar crowd crush happened in India during the Kumbh Mela Festival, where the crowd experienced sudden panic which turned tragic, ultimately resulted in sixty serious injuries and thirty fatalities [11]. Such incidents highlight the significance of modern crowd management strategies, especially in enclosed or densely populated areas. While these incidents are taken from various events, they all underscore the same underlying issue. The absence of real-time data access and crowd control, even well-planned settings can turn into a threatening event.

Crowd control is often highlighted for larger events. However, in safety critical environments, crowd control is as, if not more important and crucial to be considered seriously. Safety critical environments such as children's playgrounds, where young and vulnerable individuals are present, require extra attention for safety measures. These areas are prone to safety risks, which can lead to serious injuries or harmful incidents involving children. The demand for children's playgrounds has significantly increased over the past few years. Advanced crowd monitoring solutions have been made possible by the rapid growth of Industry 4.0 technologies. These technologies enable systems to perform and enhance real-time human detection and analyze crowd behavior for

spatial dynamics. They introduce a more intelligent approach to crowd monitoring that can generate visual representations of the data that unravel hidden crowd patterns and perform real-time monitoring alerts to notify on abnormal crowd conditions, thereby supporting informed, data-driven decision-making. Although there are already several crowd detection models available, their performance varies depending on their intended use case. Many of the existing models are developed to detect people in general, such as mask detection in malls or public surveillance. These models are trained for a certain objective using data from preset angles depending on the setting where the models are going to be deployed in. Variations in use case, camera angles, and environmental conditions limit the model's compatibility across different domains outside its intended use.

With this nature, each model must be specifically trained for a particular purpose and environment. The effort of fine-tuning object detection models to cater to a specific task has been proven to significantly improve detection accuracy [12]. Presently, very few existing models have been optimized for the purpose of human detection in indoor children's playgrounds, where smaller body sizes and frequent occlusions pose unique challenges to detection accuracy. Prior literature heavily relies on general human detection models that treat all human shapes uniformly, thereby introducing a distinct research gap: the inability to dynamically differentiate between high-dependency targets (children) and supervisors (guardians) under severe spatial occlusions and highly saturated color profiles typical of play zones. This research explicitly bridges this gap by configuring an integrated edge-to-cloud computer vision workflow optimized for highly localized playground dynamics.

The pre-trained YOLO models have been proven highly performative for fast and accurate detection, making it a suitable for real-time crowd detection. A past study proposes a child detection model built using the YOLOv5 object detection model to identify the presence of children in restricted environments [13]. The proposed model surpasses the pre-trained YOLOv5s model, conforming to the effectiveness of fine-tuning object detection models for task-specific use case. Another study conducted a comparative analysis between SSD, Faster R-CNN and various YOLO framework for the use case of people counting [14]. The YOLO family performed best in terms of detection accuracy and minimal misclassifications. Among the YOLO versions, YOLOv8 resulted in the highest overall performance, achieving accuracy of 0.92, precision of 0.90, and F1 score of 0.93. Another study proposed YOLOv5s as the best performing model for human fall detection system, achieving score for precision of 0.938, recall of 0.84, mAP@0.5 of 0.948 and mAP@0.5:0.95 of 0.588 [15]. These studies validate the superiority of YOLO models compared to the other architectures for real-time detection.

Another study integrated YOLO object detection model with IoT devices to create a real-time crowd monitoring system [16]. The findings indicate that YOLOv8NAS performed the best, achieving a mAP value of 0.951, while highlighting the potential application of a system that integrates IP cameras and computer vision for crowd monitoring. Lastly, a study proposed a real-time human detection built upon the YOLOv7-tiny architecture and trained using a custom self-collected dataset called the IHD dataset [17]. The YOLO-IHD resulted in a score of mAP@0.5

of 0.7771, surpassing the YOLOv7-tiny model which resulted in a score of mAP@0.50 of 0.352. The studies concluded that optimized YOLO models are superior in task specific use case, having balance of speed, accuracy, and computational efficiency.

The research aimed to (1) Create a new, self-collected dataset of child and guardian in an indoor playground setting, (2) Train and fine-tune four versions of the YOLO object detection models using the self-collected dataset, (3) Evaluate the performance of the trained YOLO models using standard evaluation metrics, (4) Create a crowd detection system with the trained model and integrate it with a database for storing real-time detection results, and (5) Present a method of crowd monitoring using real-time crowd data to support management decisions through real-time data visualization.

II. THEORETICAL FOUNDATION

A. Crowd Management and Smart Surveillance

Crowd management is the act of sensing and analyzing crowds to organize and influence the crowd's movement for the sake of safety and comfort for the people within that designated space [18]. The term "Crowd management" was first defined in 1993 as a proactive and systematic approach that involves planning and continuous monitoring for an orderly crowd gathering, focusing on visitor safety and well-being to avoid excessive crowd pressure and uncontrolled crowd movements [19]. Crowd monitoring system is the system utilizing technologies for the application of crowd management. These technologies can be in many forms, including sensors and analytical tools which can be used to track and analyze crowd behavior in real-time.

B. Technology Advancements and Deep Learning

IoT implies that any physical object can be connected to the internet and communicate with other objects, engaging in intelligent communication. These connecting devices, or "things", can gather data, interact with the environment, and communicate with each other to create intelligent responses [20]. The integration of these devices creates a system, enabling the automation of a specific task. An example of an IoT system is an automatic temperature detector, in which a sensor can detect the changes in temperature of a room and can automatically turn on the cooler or heater of the room depending on the situation. In the context of this thesis, the application can be in the form of an IoT based camera that can send data into an analytics system to detect the occupancy of people, or an RFID bracelet that can send information into the analytics system to show the exact location of the children.

Deep learning uses artificial neural networks (ANNs) to identify complex patterns and relationships in data. ANNs consist of multiple layers of interconnected nodes, or neurons, which enables them to learn hierarchical representations. Deep neural networks can perform tasks such as image classification, natural language processing, and object detection. However, the quality and effectiveness of its performance depend on the computing resources and datasets on which it was fed [21]. In deep learning, Convolutional Neural Networks (CNNs) are specifically designed to process grid-like data structures such as images. Image processing is possible through CNNs because it utilizes convolutional layers to identify patterns and pooling layers to reduce spatial dimensions of features while preserving essential information

[19]. Thus, CNN is suitable for object recognition, facial detection, and image segmentation [22].

You Only Look Once (YOLO) is a deep learning-based object detection model built on CNNs known for its exceptional performance on accuracy and speed which allows accurate and rapid detection of objects in images [21]. YOLO has been consistently releasing new versions with improvements on each version. A study compares the different YOLO model versions, including versions 8, 9, 10, and 11 that are used in this research [23].

The implementation of Gaussian kernels in computer vision and image processing for blurring, smoothing, and density estimation [24] can be utilized to enrich data analytics with meaningful insights, identify patterns, and draw conclusions to optimize operations and gain competitive advantages [25]. In the context of this research, data analytics can be used to identify crowd patterns, determine the most popular areas, and predict peak visitor times to improve space management and user experience, for example,

III. RESEARCH METHODOLOGY

A. Modeling Framework and Methodology

The purpose of this research is to compare and evaluate the YOLO object detection models that are fine-tuned using a custom dataset, and to integrate the best-performing model to a crowd analytics dashboard for crowd monitoring in the playground environment. Fig. 1 illustrates the research framework. The framework begins with the research objective and design, dataset collection, model development and evaluation, followed by data analytics and report writing. Dataset collection and preparation consist of manually collecting video footage, extracting footages into individual frames, ground truth annotations with bounding boxes, dataset preprocessing, and finally the splitting of dataset into 3 sets for training, validation, and testing. Model development starts with fine-tuning the YOLO models, followed by the evaluation of the model performances. The development process is the implementation of the model for a real-time playground crowd detection and crowd heatmap generation. The system then translates the result into data to be fed into the analytics dashboard for visualization. The following sections details the sequential steps undertaken in this study to achieve the intended research outcomes.

B. Dataset Collection and Preparation

A local indoor children's playground located in Sentul City was the primary place for data collection through existing CCTV system of the outlet. Out of all the available cameras, five were specifically chosen for analysis. These cameras captured the main play area clearly with minimal obstructions. This ensures that the dataset focuses solely on children's movements within the designated play area. To make sure all video footage is consistent in format, cropping, trimming, and image enhancements were conducted manually. After collection, the videos were extracted into individual image frames. The frame extraction was done using a Roboflow that sampled the video at consistent time intervals. The frames were then carefully assessed to make sure of clear visibility of objects and environment. Annotations were added to total of 650 playground images with bounding boxes around individual children and guardians to clearly define the objects for the detection task. Gathering data from a localized site imposes distinct environmental constraints. The dataset

exhibits structural bias tied to the specific architectural layout, illumination thresholds, and high-saturation color attributes of the Sentul City venue. Consequently, the models possess a localized inductive bias, meaning their generalization capability may decrease when exposed to external playground environments with varying light temperatures or alternative apparatus geometries.

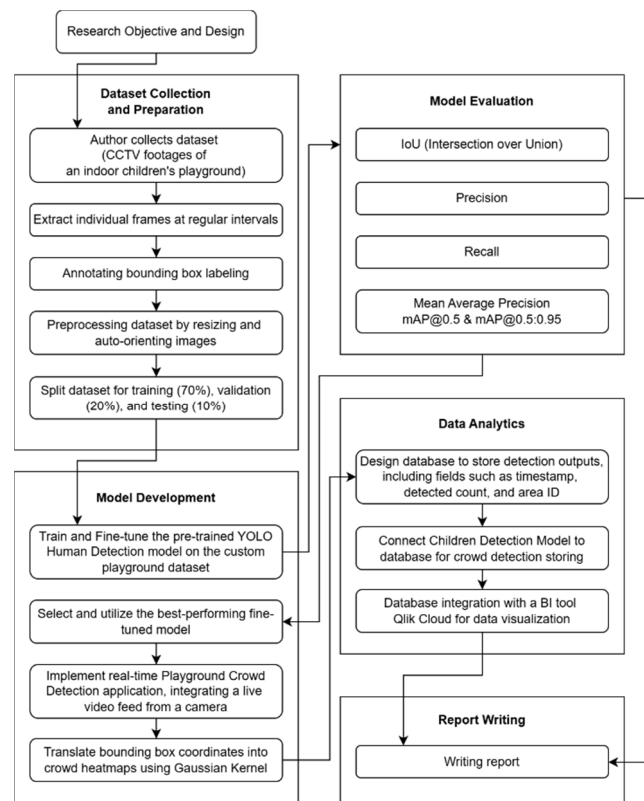


Fig. 1. Research Framework

Prior to the training, the images were resized to 640×640 pixels, which is the input size of the model. Images were also auto oriented to correct the rotations of the image that may have been incorrectly rotated. The dataset did not undergo gray scaling, for it may discard valuable chromatic information that is critical for object differentiation in a colorful environment [26]. Color variations are critical to aid the model in separating children from the background and from one another, particularly in crowded scenarios where texture and shape alone are insufficient. Fig. 2 presents the example image before and after preprocessing. After data preprocessing, the dataset was split into 3 sets. 70% was allocated for training, 20% for validation, and 10% for testing.

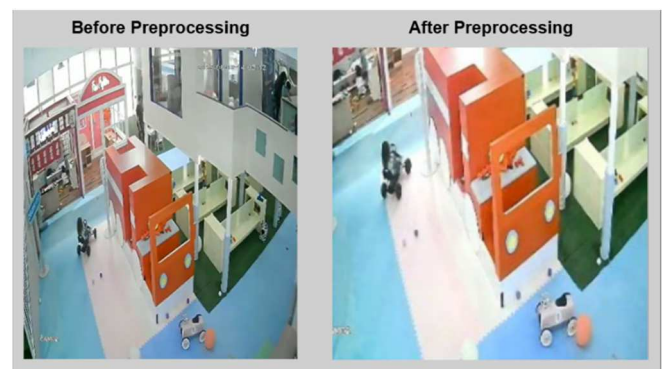


Fig. 2. Image Before and After Processing

C. YOLO Models Training

The four YOLO model versions (YOLOv8n, v9t, v10n, and v11n) were trained on the custom dataset using a consistent set of hyperparameter configurations across all model versions. The hyperparameters tuned were image size, batch size, number of epochs, and augmentation strategies. Data augmentations were applied dynamically to the images during the training process, on-the-fly, rather than modifying and storing augmented versions beforehand. The augmentations of images included flipping images horizontally, value adjustments of hue, saturation, and value, and applying mosaic augmentation.

D. Model Evaluation Metrics

The trained models were evaluated using several evaluation metrics for comparative analysis. Intersection over Union (IoU) quantifies the degree of overlap between the predicted bounding box and the ground truth bounding box, valuing between 0 and 1. Precision quantifies the proportion of correctly predicted bounding boxes among all bounding boxes predicted by the model. It reflects how many of the model's detections are true positive. Meanwhile, recall measures the proportion of the correctly identified objects to all ground truth instances. It indicates how well the model can detect all relevant instances within an image.

The Mean Average Precision (mAP) indicates the overall performance of object detection models. The value describes how well a model detects and correctly classifies objects in images. At the core of mAP is Average Precision (AP), which measures the area under the Precision-Recall (PR) curve for a single class. The mAP values the mean of all APs, providing a single score that reflects the model's overall detection performance across all classes. Both mAP@0.5 and mAP@0.5:0.95 were used in this study to evaluate the models. mAP@0.5 measures performance at a single, relatively lenient IoU threshold of 0.5, while mAP@0.5:0.95 calculates the average precision across multiple IoU thresholds ranging from 0.5 to 0.95 in increments of 0.05, providing a more rigorous and comprehensive assessment of model performance.

E. Playground Crowd Detection

The playground crowd detection system was developed to detect the presence of child and guardian from the video input. The system also generates a crowd density heatmap, counts the visible individuals within that space and inserts real-time detection input into a cloud database for data analytics. The result of the playground detection of captured child and guardian within each frame was analyzed through the application of a Gaussian kernel heatmap. The resulting heatmap provided an informative visualization of space utilization in the indoor playground, highlighting popular zones, congestion points, and potential safety concerns.

IV. RESULTS

A. Dataset Preparation and Auto-Labeling

A total of five video recordings were collected to construct the playground dataset. The recordings were carefully assessed to ensure clarity and relevance to the study. Before image extraction, the videos were cropped to a 1:1 ratio to ensure uniform sizes. Cropping was performed using local video editing software. Each video was then extracted at a one-second interval using Roboflow. Annotations were made to identify 2 classes, class Child and class Person for the guardian or adult. Roboflow's AI-assisted annotation feature,

Auto Label, was used to speed up the process of annotating the detected object. The Roboflow Grounding DINO AI model generated automatic annotations of the child and the guardian. Auto Label setting is depicted in Fig. 3.

These annotations were then reviewed one by one and refined manually to ensure accuracy. In cases where the auto-labeling results were incomplete or imprecise, additional manual annotations were added. From all the extracted images, only high-quality images with correct annotations were included in the dataset. While images containing blurred objects or vague overlaps were rejected and not included in the dataset. A total of 650 images were collected.

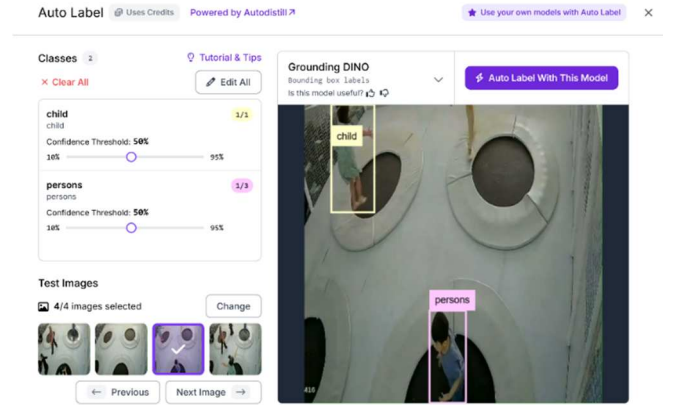


Fig. 3. Auto Label in Roboflow

During the dataset export process, parameters such as training ratio and preprocessing steps were specified. The dataset was then divided into three subsets: 70% for training, 20% for validation, and 10% for testing. From the total annotated images, 455 were allocated for training, 130 for validation, and 65 for testing. Data preprocessing consisted of applying auto-orientation to correct any image rotation during upload and resizing all images to 640 × 640 pixels.

B. Model Training

The models were trained using consistent hyperparameters across all experiments. The models were trained for 100 epochs using a batch size of 8. The `model_train()` function trains the YOLO model with the hyperparameters defined within its parameters. In addition to the evaluation metrics, training time was also measured using the `t.tic()` and `t.tocvalue()` functions. Once training was completed for all models, the results were recorded for quantitative comparison. The overall performance of the four models, including metrics values and training time, is summarized in Table I.

TABLE I. TRAINING PERFORMANCE METRICS COMPARISON

Model	P	R	mAP 50	mAP 50-90	Training time
YOLOv8	0.905	0.894	0.947	0.715	04:50:53.338
YOLOv9	0.912	0.864	0.929	0.704	06:36:47.702
YOLOv10	0.86	0.789	0.898	0.648	06:39:07.619
YOLOv11	0.928	0.854	0.939	0.703	04:29:51.179

V. DISCUSSIONS

The evaluation of the models was done by comparing the testing evaluation metrics performance. The testing metrics values are summarized in Table II. The YOLOv8 performed best overall from the training result, having the highest recall (0.894), mAP@0.50 (0.947) and mAP@0.50-0.90 (0.715).

YOLOv11 performed second, followed by YOLOv9 and YOLOv10. The YOLOv8 also performed best across all evaluation metrics during testing, followed by YOLOv9, YOLOv11, and lastly YOLOv10. Overall, YOLOv8 achieved the highest score across all evaluation metrics, achieving precision of 0.939, recall of 0.922, mAP@0.50 of 0.963 and mAP@0.50-0.90 of 0.731. Compared to the referenced studies, the result of the YOLOv8 model is comparable to the results of that research.

TABLE II. TESTING PERFORMANCE METRICS COMPARISON

Model	Precision	Recall	mAP50	mAP50-90
YOLOv8	0.939	0.922	0.963	0.731
YOLOv9	0.927	0.912	0.959	0.731
YOLOv10	0.871	0.871	0.925	0.677
YOLOv11	0.916	0.875	0.959	0.717

The four models also differ significantly in the number of parameters, capacity and computational complexity. Parameters indicate the model's learnable weights within a neural network. They determine how much information a model can store and how richly it can represent visual features. Models with more parameters generally have greater ability to learn complex patterns, but they also require more computation, longer training times, and higher risk of overfitting. Models with fewer parameters are faster and more efficient but may struggle in highly detailed or crowded scenes. YOLOv8n has the greatest number of parameters, followed by YOLOv11, YOLOv10n, and lastly YOLOv9t. YOLOv8n contains around 3.2 million parameters, giving it a balance between simplicity and representational power. Combined with its anchor-free detection head, this enables YOLOv8 to achieve the best overall performance in this study, especially in scenarios requiring both speed and accuracy. YOLOv11n, with approximately 2.6 million parameters, remains lightweight but benefits from architectural refinements that improve feature extraction efficiency, making it the second-best performer. YOLOv10 contains around 2.3 million parameters and YOLOv9 around 2.0 million parameters. YOLOv10n and YOLOv9t, having even fewer parameters and more efficiency-oriented designs, excel in simple environments but lack the capacity needed to match YOLOv8 and YOLOv11 in more complex real-world playground scenes.

The resulting playground crowd monitoring system is the integration of a crowd detection model and analytics dashboard. While the crowd detection system displays real-time crowd heatmap and occupancy count of one play area, dashboard displays real-time total count, total child count, and total guardian count of the entire playground area. The area line chart within the dashboard illustrates the occupancy trend of child and guardian over time. The blue line represents the occupancy of children, while the purple line represents the guardians. The stacked bar chart visualizes the latest occupancy distribution across different areas, where each bar represents an area, with blue indicating the child count and purple indicating the guardian. The table chart lists all areas along with their total, child, and guardian occupancy, sorted from the most crowded to the least crowded. The dashboard also displays the most crowded and least crowded areas, enabling quick identification of space utilization and crowd management status. The video from the playground crowd detection is also displayed for real-time heatmap monitoring. When a specific play area is selected, all charts dynamically

update, filtering the data to display only the information relevant to the selected area. The dashboard is also equipped with conditional formatting to display alerts, having numbers to appear red when the playground or any specific area exceeded its capacity. Fig. 4 depicts the dashboard when a specific area is selected.

Specifically for playground safety purposes, an alert system was developed to assist area keepers in monitoring potential crowding situations that may lead to accidents. The alert was made to be triggered when the child or guardian count of an area is greater than its capacity. This playground dashboard allows administrators to identify crowd patterns, monitor crowd density for safety purposes, and make data-driven decisions to optimize resource allocation and improve space management efficiency.

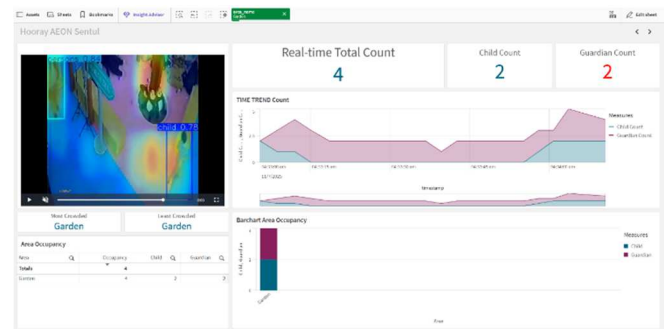


Fig. 4. Dashboard Displaying Selected Area

VI. CONCLUSIONS

This paper presents YOLOv8 as the most stable and suitable model to be used for this task. YOLOv8 performed best from the qualitative evaluation metrics from both training and testing, achieving a precision of 0.939, recall of 0.922, mAP@0.50 of 0.963 and mAP@0.50-0.90 of 0.731. The high score in evaluation metrics is reflected in the actual performance using the test set images which are the images that has not been seen by the models during training. From the visual analysis, YOLOv8 also performed well in the first image set, however not the best model within that environment. In the second image set, where complex structures and colors were present, YOLOv8 excelled in comparison to the other four models. YOLOv8 performed the most stable, having all correct detections in complex situation where the image consists of many children and guardians in one setting, simulating a crowded playground.

This result is supported by YOLOv8's architecture, which incorporates anchor-free and decoupled detection head. This architecture provides the most balance between accuracy and efficiency. Additionally, YOLOv8 has the highest number of parameters compared to the other four models, enabling it to learn richer feature representations, resulting in superior detection capability especially in diverse and complex scenarios.

The playground crowd detection system presents the possible ways of leveraging simple AI models to support business operational efficiency and improve child safety. It demonstrates a modern and practical way of monitoring children's environment, helping with better supervision and control over the area. Possible applications include applications in schools, daycares, public parks, and other spaces where children are present. The system could be further improved by enabling the model to identify additional objects,

maintain safe distances, and alert when children encounter certain areas or items.

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AUTHOR CONTRIBUTIONS

Michael Siek: conceptualize and formulate ideas, formulate methodological goal and aims, validate the system, verify the codes, revise the paper. Elizabeth Kaori Harris: identify the problems, formulate application objectives, analyse and design the system, implement program codes, test the system, write paper draft.

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